GENETICS · MCQ SET

GI Tract Genetic Disorders — Overview

LG 1
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GIT Block

Genetics GIT Block · MEDucation by Mattin Majid · Practice MCQs

HOW TO USE THIS SET

20 questions, mixed difficulty, 8 case-based. Attempt every question before checking the answer key on the last page.

4 Easy | 10 Medium | 6 Hard

1. EASY

Which gene is mutated in Familial Adenomatous Polyposis (FAP)?

A. BMPR1A
B. NOD2
C. APC
D. STK11
E. MLH1

2. EASY

Which chromosomal abnormality is classically associated with Hirschsprung disease and duodenal atresia?

A. Turner syndrome (45,X)
B. 22q11 deletion
C. Klinefelter syndrome
D. Trisomy 13
E. Down syndrome (Trisomy 21)

3. EASY

Peutz-Jeghers syndrome is characterised by hamartomatous polyps together with which additional clinical feature?

A. Mucocutaneous pigmentation of the lips and mouth
B. Aganglionic colonic segment
C. Hundreds to thousands of colonic adenomas
D. Microsatellite instability
E. Esophageal atresia

4. EASY

Which two genes are the classic somatic (non-inherited) drivers of sporadic colorectal cancer mentioned in this lecture?

A. STK11 and SMAD4
B. KRAS and TP53
C. APC and MLH1
D. HLA-DQ2 and HLA-DQ8
E. BMPR1A and NOD2

5. MEDIUM CASE

A 22-year-old undergoes colonoscopy for rectal bleeding. Hundreds of adenomatous polyps are found throughout the colon and rectum, along with several duodenal polyps.

Which gene is most likely mutated, and what is the untreated colorectal cancer risk?

A. BMPR1A; childhood-onset colorectal cancer
B. STK11; increased breast and ovarian cancer risk
C. NOD2; variable risk with fistula formation
D. APC; approximately 100% risk if untreated
E. MLH1; 3–5% of all CRC

6. MEDIUM CASE

A 45-year-old woman with only a handful of colonic adenomas develops colorectal cancer. Her mother had endometrial cancer.

Which molecular mechanism best explains this pattern?

A. Microsatellite instability from a mismatch repair (MMR) gene mutation
B. A somatic KRAS mutation only
C. Trisomy of chromosome 21
D. TGF-β pathway disruption
E. Chromosomal nondisjunction

7. MEDIUM

Which feature most reliably distinguishes Lynch syndrome from FAP?

A. Lynch syndrome is a chromosomal, not single-gene, disorder
B. Lynch syndrome causes few or no polyps despite high cancer risk, unlike the hundreds to thousands seen in FAP
C. Lynch syndrome has more polyps than FAP
D. Lynch syndrome only affects the stomach
E. FAP has a lower lifetime cancer risk than Lynch syndrome

8. MEDIUM CASE

A newborn with Down syndrome presents with bilious vomiting and a 'double bubble' sign on abdominal X-ray.

What is the diagnosis?

A. Pyloric stenosis
B. Imperforate anus
C. Duodenal atresia
D. Esophageal atresia
E. Hirschsprung disease

9. MEDIUM CASE

Colon biopsy shows transmural, patchy inflammation with skip lesions and a fistula tract.

Which inflammatory bowel disease subtype is this?

A. Microscopic colitis
B. Ulcerative colitis
C. Crohn's disease
D. Celiac disease
E. Ischaemic colitis

10. MEDIUM

Which gene is associated with both Crohn's disease and ulcerative colitis, reflecting their shared polygenic basis?

A. HLA-DQ2
B. NOD2
C. TYMP
D. APC
E. STK11

11. MEDIUM

Celiac disease is mechanistically driven by which process in genetically susceptible individuals carrying HLA-DQ2/DQ8?

A. Gluten ingestion triggering immune-mediated destruction of small intestinal villi
B. A somatic KRAS mutation in enterocytes
C. Chromosomal nondisjunction affecting intestinal cells
D. Mitochondrial DNA depletion in villous epithelium
E. Loss of the APC tumour suppressor

12. MEDIUM

Untreated celiac disease carries a long-term increased risk of which malignancy?

A. Small bowel lymphoma

B. Colorectal cancer

C. Hepatocellular carcinoma

D. Pancreatic cancer

E. Gastric adenocarcinoma exclusively

13. MEDIUM CASE

A teenager presents with dark pigmented macules around the mouth and is found to have hamartomatous polyps concentrated in the small intestine.

Which gene is implicated in this syndrome?

A. NOD2

B. MSH2

C. APC

D. STK11 (LKB1)

E. BMPR1A

14. MEDIUM

Juvenile polyposis syndrome, caused by mutations in the TGF-β pathway genes BMPR1A or SMAD4, is best distinguished from FAP by which feature?

A. It exclusively affects the duodenum

B. It causes thousands of adenomatous polyps

C. It has a 100% penetrance for CRC by adulthood

D. It presents with hamartomatous polyps in childhood, primarily in the colon

E. It is caused by an APC mutation

15. HARD CASE

A patient is diagnosed with Mitochondrial Neurogastrointestinal Encephalomyopathy (MNGIE), presenting with severe GI dysmotility and peripheral neuropathy.

Which gene/enzyme deficiency underlies this condition, and what is the consequence?

A. HLA-DQ2 expression causing autoimmune enteropathy

B. MLH1 deficiency causing microsatellite instability

C. TYMP (thymidine phosphorylase) deficiency causing toxic nucleoside buildup and mitochondrial damage

D. APC deficiency causing polyposis

E. NOD2 mutation causing granulomatous inflammation

16. HARD

A colorectal cancer arises with no family history, showing KRAS and TP53 mutations acquired only within the tumour tissue. This is best classified as:

A. A mitochondrial inheritance pattern

B. An HLA-linked autoimmune process

C. A sporadic cancer driven by somatic mutation, not inherited

D. A chromosomal abnormality syndrome

E. An inherited polyposis syndrome

17. HARD CASE

A 35-year-old woman with Lynch syndrome (confirmed MSH2 mutation) asks about her risk beyond colorectal cancer.

Which additional cancers are most relevant to counsel her about, based on Lynch syndrome's known associations?

A. No increased risk beyond the colon

B. Endometrial, gastric, and ovarian cancer

C. Pancreatic cancer exclusively

D. Breast and ovarian cancer only, as in Peutz-Jeghers

E. Only skin cancers, as in juvenile polyposis

18. HARD

Compared to Down syndrome, Turner syndrome (45,X) is associated with which GI manifestations, according to this lecture?

A. Pyloric stenosis and inflammatory bowel disease, less commonly than Down syndrome's GI anomalies

B. Duodenal atresia and imperforate anus, identical to Down syndrome

C. No GI manifestations are described

D. Hirschsprung disease exclusively

E. Esophageal atresia as the dominant feature

19. HARD CASE

A pathologist reviewing a colorectal cancer specimen notes abnormal DNA methylation silencing a tumour suppressor gene, with no underlying germline mutation identified in the patient.

This mechanism is best classified as:

A. An epigenetic (non-inherited) change driving sporadic cancer

B. An HLA-associated autoimmune mechanism

C. A mitochondrial DNA mutation

D. A chromosomal trisomy

E. A single-gene inherited mutation

20. HARD

A patient has hundreds of colonic polyps beginning in adolescence, plus desmoid tumours. Genetic testing is pending. Which finding would most strongly argue AGAINST a diagnosis of Lynch syndrome and FOR FAP?

A. Evidence of microsatellite instability on tumour testing

B. A mismatch repair gene mutation identified on panel testing

C. Isolated gastric cancer without any colonic polyps

D. The sheer polyp burden (hundreds to thousands) and early adolescent onset, which are characteristic of FAP rather than the low polyp count of Lynch syndrome

E. A family history of endometrial cancer

ANSWER KEY & RATIONALE

Don't peek until you've committed to an answer on every question.

Q	Ans	Why
1	**C**	APC
2	**E**	Down syndrome (Trisomy 21)
3	**A**	Mucocutaneous pigmentation of the lips and mouth
4	**B**	KRAS and TP53
5	**D**	APC; approximately 100% risk if untreated
6	**A**	Microsatellite instability from a mismatch repair (MMR) gene mutation
7	**B**	Lynch syndrome causes few or no polyps despite high cancer risk, unlike the hundreds to thousands seen in FAP
8	**C**	Duodenal atresia
9	**C**	Crohn's disease
10	**B**	NOD2
11	**A**	Gluten ingestion triggering immune-mediated destruction of small intestinal villi
12	**A**	Small bowel lymphoma
13	**D**	STK11 (LKB1)
14	**D**	It presents with hamartomatous polyps in childhood, primarily in the colon
15	**C**	TYMP (thymidine phosphorylase) deficiency causing toxic nucleoside buildup and mitochondrial damage
16	**C**	A sporadic cancer driven by somatic mutation, not inherited
17	**B**	Endometrial, gastric, and ovarian cancer
18	**A**	Pyloric stenosis and inflammatory bowel disease, less commonly than Down syndrome's GI anomalies
19	**A**	An epigenetic (non-inherited) change driving sporadic cancer
20	**D**	The sheer polyp burden (hundreds to thousands) and early adolescent onset, which are characteristic of FAP rather than the low polyp count of Lynch syndrome